

# Caleb Adams

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LinkedIn: [Caleb Adams](#) | Portfolio: [Engineering Portfolio](#)

## PROFESSIONAL SUMMARY

Mechanical engineering student with hands-on experience in HVAC installation, CAD design, prototyping, machining, and embedded systems. Passionate about product design, manufacturing, and solving real world engineering problems through iterative design and testing.

## EDUCATION

**The Ohio State University:** Columbus, Ohio

**Bachelor of Science in Mechanical Engineering, Business Minor,** Expected May 2028

**Relevant Courses:** Linear Algebra and Differential Equations, Thermodynamics, Dynamics, Mechanics of Materials, Statics, Numerical Methods, Calculus 1-3, Physics I, II, Material Science.

## EXPERIENCE

**Adams Heating and Cooling**

Cincinnati, OH

**HVAC Intern**

May 2025-August 2025, May 2026-August 2026

- Installed and serviced residential and commercial HVAC systems, including AC units, heat pumps, and mini splits.
- Fabricated refrigerant piping using pressing and bending equipment to minimize oxidation and eliminate debris.
- Performed nitrogen pressure testing, vacuum evacuation, and refrigerant charging.
- Completed electrical connections from disconnect boxes to unit, including 460 V three phase systems.
- Operated HVAC installation equipment, sheet metal fabrication tools, power tools safely and efficiently.

## PROJECTS

**Machine Piston**

January 2026-March 2026

- Manufactured a working aluminum-brass piston assembly using manual lathes and milling machines maintaining a tolerance of  $\pm 0.001$  in.
- Used techniques such as facing, reaming, and threading to shape the piston block, connecting rod, and head.
- Interpreted engineering drawings and used dimension analysis throughout the machining process.
- Assembled and functionally tested the completed piston assembly, achieving smooth operation at 320 RPM.

**Project Sleep: Smart Sleeping Device**

January 2025-May 2025

- Designed CAD models and developed a sleep enhancing device using MATLAB and Arduino.
- Fabricated and tested multiple prototypes, reducing size by 40% after user feedback.
- Integrated electronic components and validated system functionality through prototype testing.
- Presented at the Ohio State University Engineering Design Competition, where it ranked top 20% of 200+ teams.

**Arduino Musical Instrument Controller**

March 2026-May 2026

- Designed and built a sound system that allows the user to play 4 instruments on an IR remote.
- Integrated LCD screen, IR receiver, MP3 player module, and speakers to create an interactive instrument.
- Developed C++ code to process remote inputs, audio files, and customize the instrument. Millis function used to eliminate playback latency and improve real time responsiveness.
- Performed circuit analysis, component testing, and debugging to ensure correct use.

**3D Printed F1 Toy Car**

February 2024-May 2024

- Designed and built a battery powered toy F1 car using CAD models (Fusion 360) and 3D printing.
- Integrated electrical circuits, motors, 9V battery, and fan propulsion into the design using soldering.
- Developed an initial cardboard prototype and improved the design through iterative changes after testing.

## SKILLS

**Technical Skills:** CAD (Onshape, Fusion 360), HVAC Installation Equipment, Power Tools, Soldering, Brazing, Bending, Pressing, Manual Lathe, Milling Machine, Reaming, Threading, Facing, 3D Printing, MATLAB, C++, QuickBooks, Excel.

## EXTRACURRICULARS

**American Society of Mechanical Engineers  
Member**

Columbus, OH

August 2025-Present

- Participated in networking events and opportunities with engineering professionals.